EXP5: Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

```
clc;
clear;
close all;

f0 = 10;          % Design frequency (GHz)
Z0 = 50;          % Reference impedance (ohms)
fstart = 8;       % Start frequency (GHz)
fstop = 12;       % Stop frequency (GHz)
Ctarget = 10;     % Coupling target (dB)

f = linspace(fstart, fstop, 401);

% Directional Coupler S-parameters
S11 = -25 - 3*((f - f0)/2).^2;
S21 = -0.5 - 0.1*(f - f0).^2;
S31 = -Ctarget - 0.5*(f - f0).^2;
S41 = -28 - 2*cos((f - f0)*pi/2);

% Calculate Coupler Parameters
Coupling = -S31;
Isolation = -S41;
InsertionLoss = -S21;
Directivity = Isolation - Coupling;

% Wilkinson Power Divider S-parameters
S11_div = -25 - 2*((f - f0)/2).^2;
S21_div = -3.01 - 0.1*(f - f0).^2;
S31_div = -3.01 - 0.1*(f - f0).^2;
S23_div = -30*ones(size(f));

% Power Division Ratio
P21 = 10.^(S21_div/10);
P31 = 10.^(S31_div/10);
PowerRatio = P21 ./ P31;

% Find frequency index at 10 GHz
[~, idx] = min(abs(f - f0));

% Display Results
fprintf('\nEXPERIMENT 5 RESULTS AT %.2f GHz\n', f(idx));
fprintf('Reference impedance = %.0f ohms\n', Z0);
fprintf('Frequency sweep = %.1f to %.1f GHz\n', fstart, fstop);

fprintf('\nDIRECTIONAL COUPLER\n');
fprintf('S11 = %.2f dB\n', S11(idx));
fprintf('S21 = %.2f dB\n', S21(idx));
fprintf('S31 = %.2f dB\n', S31(idx));
fprintf('S41 (isolated port) = %.2f dB\n', S41(idx));
fprintf('Coupling = %.2f dB (target 10 dB)\n', Coupling(idx));
fprintf('Isolation = %.2f dB (target >= 25 dB)\n', Isolation(idx));
fprintf('Insertion loss = %.2f dB (target <= 1.5 dB)\n', InsertionLoss(idx));
fprintf('Directivity = %.2f dB (target >= 15 dB)\n', Directivity(idx));
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fprintf('\nTWO-WAY POWER DIVIDER\n');
fprintf('S11 = %.2f dB\n', S11_div(idx));
fprintf('S21 = %.2f dB (target -3.01 dB)\n', S21_div(idx));
fprintf('S31 = %.2f dB (target -3.01 dB)\n', S31_div(idx));
fprintf('Output isolation S23 = %.2f dB\n', S23_div(idx));
fprintf('Power division ratio P21/P31 = %.3f:1\n', PowerRatio(idx));

% Plot Results
figure('Name', 'Experiment 5 - Directional Coupler and Power Divider', ...
    'NumberTitle', 'off', 'Color', 'w');

subplot(3,4,1);
plot(f,S11,'LineWidth',1.5);
grid on;
title('Coupler S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3,4,2);
plot(f,S21,'LineWidth',1.5);
grid on;
title('Coupler S21');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3,4,3);
plot(f,S31,'LineWidth',1.5);
grid on;
title('Coupler S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3,4,4);
plot(f,S41,'LineWidth',1.5);
grid on;
title('Coupler S41 - Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3,4,5);
plot(f,Coupling,'b','LineWidth',1.5);
hold on;
yline(Ctarget,'r--','10 dB Target');
grid on;
title('Coupling');
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');

subplot(3,4,6);
plot(f,Isolation,'b','LineWidth',1.5);
hold on;
yline(25,'r--','25 dB Target');
grid on;
title('Isolation');
xlabel('Frequency (GHz)');
ylabel('Isolation (dB)');
```

```
subplot(3,4,7);
plot(f,InsertionLoss,'b','LineWidth',1.5);
hold on;
yline(1.5,'r--','1.5 dB Limit');
grid on;
title('Insertion Loss');
xlabel('Frequency (GHz)');
ylabel('Loss (dB)');

subplot(3,4,8);
plot(f,Directivity,'b','LineWidth',1.5);
hold on;
yline(15,'r--','15 dB Target');
grid on;
title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');

subplot(3,4,9);
plot(f,S11_div,'LineWidth',1.5);
grid on;
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

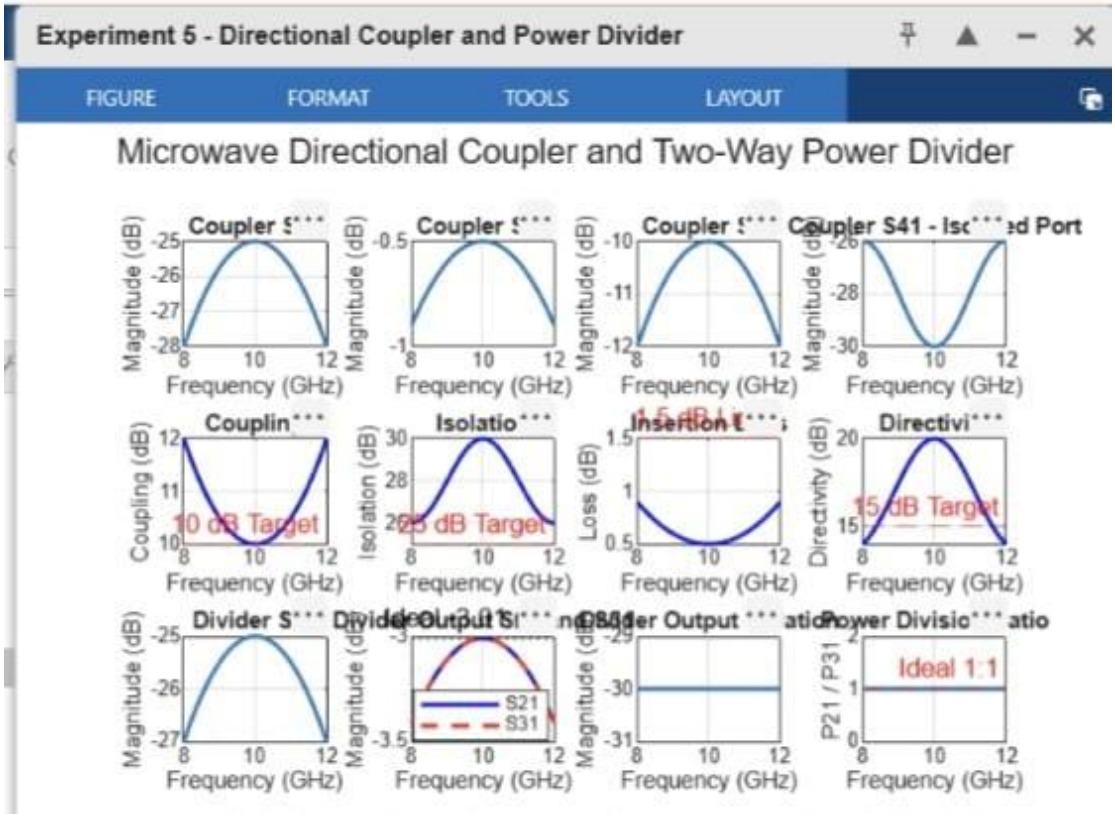
subplot(3,4,10);
plot(f,S21_div,'b',f,S31_div,'r--','LineWidth',1.5);
hold on;
yline(-3.01,'k:','Ideal -3.01 dB');
grid on;
title('Divider Output S21 and S31');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
legend('S21','S31','Location','best');

subplot(3,4,11);
plot(f,S23_div,'LineWidth',1.5);
grid on;
title('Divider Output Isolation');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3,4,12);
plot(f,PowerRatio,'LineWidth',1.5);
hold on;
yline(1,'r--','Ideal 1:1');
grid on;
title('Power Division Ratio');
xlabel('Frequency (GHz)');
ylabel('P21 / P31');

sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

OUTPUT:


Experiment 5 - Directional Coupler and Power Divider
FIGURE
FORMAT
TOOLS
LAYOUT
Microwave Directional Coupler and Two-Way Power Divider
Coupler S41 - Isc
ed Port
Couplin
Isolatio
Directivi
10 dB Target
15 dB Target
Divider S
Power Divisio
atio
Ideal 1:1
S21
S31
Magnitude (dB)
Coupling (dB)
Isolation (dB)
Loss (dB)
Directivity (dB)
P21 / P31
Frequency (GHz)